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A comparative study of the evolution of a close binary using a standard and an improved technique for computing mass transfer

U. Kolb and H. Ritter

Max-Planck-Institut für Physik und Astrophysik, Karl-Schwarzschild-Strasse 1, D-8046 Garching, Federal Republic of Germany

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Abstract. We compute the evolution of a close binary with initial parameters $M_1 = 2\ M_\odot$, $M_2 = 1\ M_\odot$ and $a = 8\ R_\odot$ using a standard technique (Kippenhahn et al., 1967) and a much more elaborate one for computing mass transfer in order to check to what extent the results differ. In the improved technique, the radius of the donor is not constrained by its critical Roche radius. Rather, the mass transfer rate derived in this paper is given as an explicit function of known stellar and binary parameters, whereby the finite atmospheric scale height and the detailed structure of the outer layers of the donor are taken properly into account. Numerical computations for this particular binary carried out by using both techniques do not yield significantly different results. Based on this and on the results of previous studies where the standard technique has been shown to fail (Pastetter and Ritter, 1989) or to yield unsatisfactory results (D'Antona et al., 1989), we draw the following, more general conclusions: The standard technique for computing mass transfer yields acceptable results if (1) the mass transfer rate is a sensitive function of the difference between the stellar radius and the corresponding critical Roche radius, (2) the density/pressure scale height H in the outer layers of the donor is small compared to its radius R, i.e. $H/R \ll 1$, and (3) the phases of turning on or off mass transfer are not important.

Key words: close binaries – stellar evolution – mass transfer

1. Introduction

It is now almost a quarter of a century since the first detailed numerical studies of the evolution of close binary systems were performed (Paczyński, 1966; Kippenhahn and Weigert, 1967). Since then numerous studies have been devoted to this subject (for reviews see e.g. Thomas, 1977; Webbink, 1979, 1985). In the terminology introduced by Kippenhahn and Weigert (1967), most model computations dealt with either a case A or case B mass transfer, while a third mode, case C, introduced by Lauterborn (1970) found comparatively little attention. The main reason for this is that unless very special initial conditions are chosen, mass transfer in a case C evolution becomes dynamically (or convectively) unstable. Recently, however, Pastetter and Ritter (1989), hereafter PR, have thoroughly discussed case C evolution (including the important effects of a stellar wind) and presented numerical computations of some special cases. One of the major (and unexpected) results of this study was the following: if mass loss occurs from thermally pulsing stars (on the asymptotic giant branch, AGB) and if this mass loss is computed by using the standard technique applied in the aforementioned papers, the results are qualitatively incorrect.

The standard technique, as we shall hereafter call it, assumes that at all time during the mass transfer phase the following two conditions are fulfilled and, in fact, may be used to determine the mass transfer rate:

$$R = R_R \tag{1}$$

and

$$\dot{R} = \dot{R}_R. \tag{2}$$

Here R and R_R are respectively the radius and the critical Roche radius of the mass losing star (hereafter the primary). As was shown e.g. by Ritter (1988), Eq. (2) holds only if mass transfer is (quasi-) stationary, i.e. if $\ddot{M}_1 \approx 0$, where M_1 denotes the mass of the primary. On the other hand, Eq. (1) implies implicitly that the mass losing star has a sharp stellar rim. Whenever this is not a good approximation, as in the examples discussed by PR, or if mass transfer is not nearly stationary, as e.g. at the turn-on of mass transfer (Ritter, 1988; D'Antona, Mazzitelli and Ritter, 1989, hereafter DMR, and Savonije, 1979, in a slightly different way), or during a thermal pulse of the donor (see PR), Eqs. (1) and (2) must not be used.

In the light of all this, therefore, the question arises to what extent the results of the numerous classical studies of binary evolution still hold. The answer is of course already known in the extreme limits: Eqs. (1) and (2) yield acceptable results if the rim of the donor star is sufficiently sharply defined, i.e. if the ratio of atmospheric pressure scale height H_p to stellar radius R of the donor is $H_p/R \ll 1$. This was shown e.g. by DMR. On the other hand, Eqs. (1) and (2) fail if H_p/R is not small, as was shown by PR. What is not known so far (because to the best of our knowledge the question has never been addressed before) is whether Eqs. (1) and (2) are acceptable in all those numerous classical studies of binary evolution where they have been used, i.e. if mass loss occurs from subgiants or giants, that is from stars that are more extended than main sequence stars but less extended than stars on the AGB.

With this end in view, we have recomputed the classical low-mass case B evolution first studied by Kippenhahn, Kohl and Weigert (1967), hereafter KKW, which involves mass loss from a subgiant/giant donor. For this we use, on the one hand, the

Send offprint requests to: U. Kolb

standard technique, i.e. Eqs. (1) and (2), and on other a much more elaborate one (in the spirit of PR and DMR) for computing the mass transfer rate. In this way, our study is strictly differential in the sense that all differences in the results of the two model sequences must be attributed to the different techniques used to determine $\dot{M}$.

During the mass transfer phase of this evolution the donor may overfill its critical Roche lobe. Since in this case the optically thin approximation for computing $\dot{M}$ used in PR and DMR is invalid and the optically thick approximations given by Paczyński and Sienkiewicz (1972) or Savonije (1978) are inadequate for the donor in question, we have elaborated on the methods by Savonije (1978) in order to obtain a physically reasonable prescription for computing $\dot{M}$. Our method will be outlined in Sect. 2, whereas details are given in an Appendix. In Sect. 3 we describe briefly the stellar evolution code and the input physics we used. In Sect. 4 we present our numerical results. A discussion and the conclusions of this paper follow in Sects. 5 and 6, respectively.

2. Computation of the mass transfer rate

In this section we wish to give a brief outline of how we compute the mass transfer rate without making use of Eqs. (1) and (2). Our aim is to obtain an expression for $\dot{M}$ as a function of known stellar and binary parameters. For this we elaborate on previous work, mainly by Paczyński and Sienkiewicz (1972), Savonije (1978), and Ritter (1988).

Starting point is a simplified hydrodynamical model of mass transfer as discussed by Lubow and Shu (1975). Accordingly, the flow of gas is stationary and subsonic with streamlines remaining on constant (Roche) potential Φ. The flow accelerates and converges towards the inner Lagrangian point L_1 and reaches sound velocity near the L_1-plane (this is the plane perpendicular to the line joining the two stars and going through L_1). In this model only the outermost layers of the donor, i.e. those at a potential $\Phi > \Phi_L$, where Φ_L is the critical Roche potential of L_1, are affected by the mass flow. Layers at $\Phi < \Phi_L$ remain at rest. Accordingly the mass transfer rate is then obtained by integrating the mass flux density over the L_1-plane F:

$$-\dot{M}_1 = \dot{M} = \int_F \rho_L v_s \mathrm{d}A. \tag{3}$$

Here ρ_L and v_s are respectively the density and the sound speed of the flow through the L_1-plane. In order to relate the a priori unknown quantities in the L_1-plane to known quantities of a (1-dimensional) hydrostatic stellar model of the primary we make use of Bernoulli's equation. This equation relates the properties of a stationary laminar flow at two arbitrary points on a stream line. Here we chose one point in the L_1-plane and the other, hereafter denoted by E, somewhere on the primary (far) away from L_1. In the following we shall denote quantities referring to point E with a subscript 0. Accordingly

$$\frac{1}{2}v_0^2 + \Phi_0 = \int_E^L \frac{\mathrm{d}P}{\rho} + \frac{1}{2}v_L^2 + \Phi_L. \tag{4}$$

Now we chose point E sufficiently far away from L_1 where the flow velocity is small, i.e. $v_0^2 \ll v_s^2$. At such a point the vertical stratification of the donor is in hydrostatic equilibrium and well approximated by the structure of the corresponding 1-dimensional stellar model in hydrostatic equilibrium. Because in our model $v_L = v_s$ and $\Phi_0 = \Phi_L$, Eq. (4) simplifies considerably to

$$\int_E^L \frac{\mathrm{d}P}{\rho} + \frac{1}{2}v_s^2 = 0. \tag{5}$$

Since in the context of this paper, which deals with mass transfer on a thermal or nuclear time scale, we wish to avoid a detailed hydrodynamical treatment of the flow, we introduce one further simplification in order to evaluate the integral in Eq. (5). For this we subdivide the flow in an optically thin and an optically thick part and solve for its structure separately.

In the optically thin part (optical depth $\tau < \tau_{ph} \approx 2/3$) the flow occurs from above the photosphere. Because of the low optical depth, we assume the flow to be isothermal at the effective temperature of the atmosphere, i.e. $T = T_{eff} = \mathrm{const}$. The resulting equation for the rate of optically thin mass transfer is given in the Appendix [Eqs. (A1)–(A5)]. A detailed derivation has been given by Ritter (1988). In the Appendix we describe also an algorithm which avoids expanding the Roche potential in terms of the volume radius used by Ritter (1988).

In the optically thick layers (here where $\tau > \tau_{ph}$) we assume the flow to be adiabatic, i.e. along a stream line

$$P = K\rho^{\Gamma_1}, \tag{6}$$

where K and the adiabatic exponent $\Gamma_1 = (\partial \ln P/\partial \ln \rho)_{ad}$ are constant. It is, however, clear that in general Γ_1 does change along a stream line because of the varying degree of ionisation of the gas.

Using Eqs. (5) and (6) it is possible to rewrite the integral in Eq. (3) as one over the outermost layers (with $\Phi_L < \Phi < \Phi_{ph}$) of a 1-dim. stellar model of the primary. A detailed derivation is given in the Appendix.

We wish to end this section with a brief discussion of those points in which our approach differs from previous similar ones. The main difference to Paczyński and Sienkiewicz (1972), Plavec et al. (1973) and Savonije (1978) is that we do not use a polytropic approximation for the radial structure of the layers with $\Phi_L < \Phi < \Phi_{ph}$ in order to arrive at an analytical solution of Eq. (3). Rather we take the vertical structure of the primary as computed by the stellar evolution code and perform the transformed integral in Eq. (A17) numerically. This approach allows us to handle mass loss from stars that have a density inversion in their outer layers and for which a polytropic approximation is not possible. Furthermore, we must reemphasize that in the framework of this model one must distinguish between the polytropic index $n_1 = 1/(\Gamma_1 - 1)$ along a streamline of the optically thick flow [Eq. (6)] and the (local) polytropic index n_2 of the vertical stratification of the donor. We note, in particular, that the result given by Savonije (1978) assumes implicitly $n_1 = n_2 = 3/2$. Therefore his equation for $\dot{M}$ holds strictly only in the case of mass loss from a convective envelope with $\nabla_a = (\partial \ln T/\partial \ln P)_{ad} = 0.4$. Finally we mention that Paczyński and Sienkiewicz (1972) as well as Savonije (1978) do not take into account the contribution from optically thin mass transfer as we do, although this mode of mass loss may be dominant, as we shall see below.

3. The stellar evolution program, input physics and model assumptions

We use an updated version of the stellar evolution code described by Kippenhahn et al. (1967) with some improvements in the input physics as described by Weiss (1986, 1987) and in PR. In addition the program has been supplemented by the subroutines for computing the mass transfer rate according to the equations given in the Appendix.

The initial chemical composition of the primary is $X=0.739$, $Y=0.240$ and $Z=0.021$ (in obvious notation). Where convection is overadiabatic we chose the ratio of mixing length to pressure scale height to be 1.5.

The binary evolution is computed by assuming conservative mass transfer, i.e. with constant total mass and orbital angular momentum. Only the evolution of the donor is followed. The secondary acts as a point mass. It is assumed to accept all the mass transferred and not to interfere with the mass transfer process.

4. Numerical computations

4.1. The initial binary system

For the purpose of this study we have chosen to follow the evolution of a binary system as similar as possible to the one computed by KKW. There are a number of reasons which led to this choice:

(a) The low-mass case B evolution in question involves mass loss from a subgiant and in later phases from a giant primary and is, therefore, appropriate for our purpose.

(b) By following the KKW system, our computations done by using the standard technique for computing $\dot{M}$ will show to what extent changes introduced in the evolutionary code, the input physics (opacities, initial chemical composition) change the outcome of the evolution.

(c) Since the evolution computed by KKW is explained in great detail, we may concentrate here on those aspects which are of particular relevance for this study and refer the reader to KKW for further details.

(d) Low-mass case B evolution, in particular the phases where mass transfer is driven by nuclear evolution of the donor, has recently become of new interest in the context of the evolution/formation of certain binary millisecond pulsar systems, as e.g. PSR 1855+09 (Segelstein et al., 1986), PSR 1953+29 (Boriakoff et al., 1983), and PSR 1620–26 (Lyne et al., 1988). Our results thus check also the validity of simplified numerical models which used Eqs. (1) and (2) to compute $\dot{M}$ (e.g. Webbink et al., 1983; Taam, 1983).

The initial parameters of the binary system we studied are listed in Table 1. Comparing them with those used by KKW it is seen that the orbital separation is slightly larger in our case. This is entirely a consequence of the higher initial hydrogen content, $X=0.739$ (as compared to $X=0.602$ in KKW), which in turn yields a slightly more extended star after central hydrogen burning. In order to initiate mass transfer at an evolutionary state of the primary equivalent to that in the KKW model, we had to choose an initial separation of 8 $R_\odot$ (as compared to 6.6 $R_\odot$ in KKW).

We have followed the evolution of this binary system in two separate sequences. In sequence A we use the relations given in the Appendix to compute $\dot{M}$. In sequence B we used the algorithm outlined in Kippenhahn and Weigert (1967), i.e. the standard technique to compute mass transfer.

Table 1. Initial and final stellar and binary parameters of the system studied in Sect. 4. M_1, M_2, $q=M_1/M_2$, a and P are respectively the masses of the primary (the donor), of the secondary, the mass ratio, the orbital separation and the orbital period

Age (10^6 yr)	$M_1/M_\odot$	$M_2/M_\odot$	$a/R_\odot$	P (d)	Points
0–1207	2.00	1.00	8.0	1.51	A, . . . , D
1406	0.27	2.73	59.1	30.4	K

4.2. The evolution in the HRD

Figure 1 shows the evolutionary track of the primary in the HRD. The full line is our computation (sequence B) whereas the dashed one is the track computed by KKW. As can be seen the results are in fact very similar, which allows us to keep the following description of the evolution very brief and to refer the reader to KKW for further details.

Now, Fig. 2 shows the evolution of the primary in the HRD for both sequences A (full line) and B (dashed line where it differs

Table 2. Explanation of important points labelled in Figs. 2, 3, 4, 5 and 6

A	ZAMS, $M_1=2.0\ M_\odot$, $X=0.739$, $Z=0.021$
C	Exhaustion of central hydrogen burning
D	Onset of the optically thick mass transfer, $q=2$, $a=8.0\ R_\odot$
U1	$M_1=M_2$, i.e. $q=1$
E	Minimum value of the Roche radius R_R
U2	End of the phase of optically thick mass transfer
U4	Deepest penetration of the outer convection zone
G	Hydrogen shell source reaches a step in the H-profile, ⇒ system is detached until point I
I	Onset of the second phase of mass transfer ($\dot{M}\geq 10^{-10}\ M_\odot\ \mathrm{yr}^{-1}$)
K	End of the mass transfer phase ($\dot{M}\leq 10^{-10} M_\odot\ \mathrm{yr}^{-1}$)

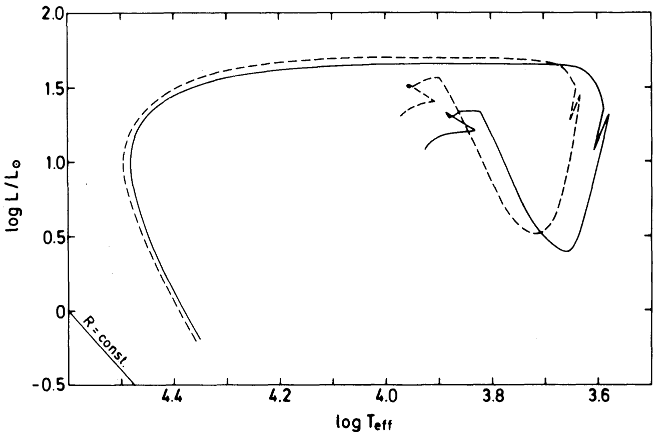

Fig. 1. HRD showing the evolutionary track of the primary in sequence B (solid line) and for a similar system (dashed) computed by Kippenhahn et al. (1967)

sufficiently from A to be shown). The evolution of the primary begins on the zero age main sequence (ZAMS, point A in Fig. 2). The critical Roche radius is reached for the first time during shell hydrogen burning (at point D in sequences A and B). At the end of the mass transfer phase (at point K, where the hydrogen shell source is still burning) the star has lost essentially its whole hydrogen-rich envelope. The remnant primary has a mass of only 0.27 $M_\odot$, 96% of which are in the degenerate helium core and 4% in the very extended and diluted hydrogen-rich envelope. The final orbital parameters resulting from that evolution are also listed in Table 1.

4.3. *The mass transfer*

Figures 3 and 4 show respectively for sequences A and B the run of the mass transfer rate with time. The irregularities seen in these curves reflect numerical noise and have no physical meaning. Despite that we refrained from showing smoothed results in order

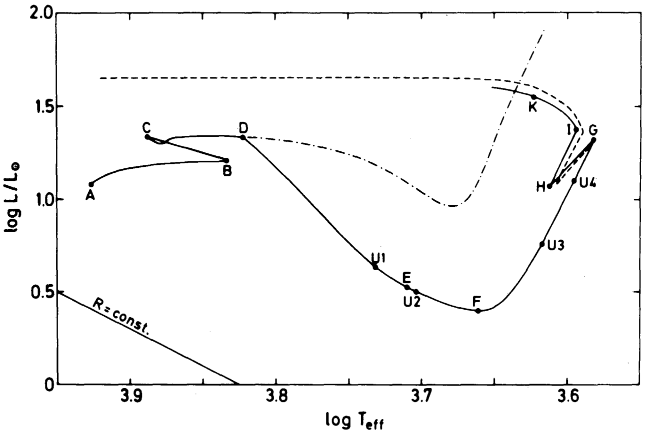

Fig. 2. Hertzsprung-Russel-diagram (HRD). The solid line shows the evolutionary track of the primary of sequence A (new method). The points labelled by letters are referred to in the text and in Table 1. Their meaning is explained in Table 2. The dashed line represents the evolutionary track for sequence B (standard technique); up to point G there is no difference between both sequences on the scale of this diagram. The evolutionary path of a 2.0 $M_\odot$ single star is also shown (dash-dotted). The straight line in the lower left is a line of constant stellar radius

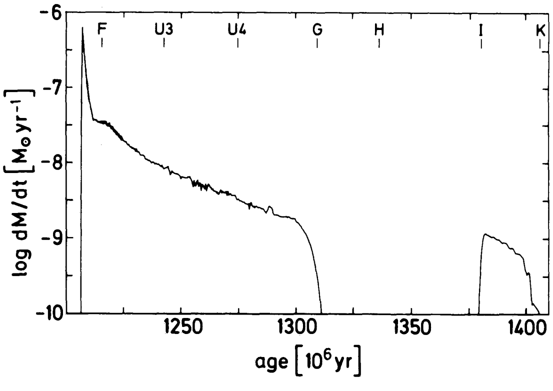

Fig. 3. The mass transfer rate $\dot{M}$ as a function of time in sequence A (new method). The labels on the top indicate characteristic evolutionary stages and are explained in Table 2

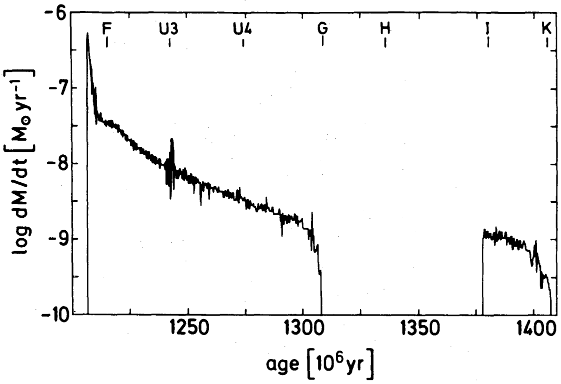

Fig. 4. The same as Fig. 3 but for sequence B (standard technique)

to give an idea of the numerical reliability and of the difficulties associated with the two methods used to compute $\dot{M}$. We shall come back to these problems in Sect. 5.3. As before in Fig. 2, we find agian only very small differences when comparing sequence A with B, i.e. Fig. 3 with Fig. 4. The only difference worth mentioning is the finite slope of $\dot{M}(t)$ during the turning on/off of mass transfer (e.g. near points G and I in Fig. 3) obtained with the new method. This is, however, not surprising. Whereas in sequence B (standard technique) we expect $\dot{M}(t)$ to be discontinuous (because of the implied sharp stellar rim), i.e. jumping from a finite value to zero and vice versa, the optically thin mass loss from the atmosphere allowed for in sequence A results in a gradual turn on/off of mass transfer.

Figure 5 shows the difference $\Delta R = R - R_\mathrm{R}$ in sequence A in units of

$$H_\mathrm{p} = \frac{\mathcal{R} T_\mathrm{eff} R_\mathrm{R}^2}{G M_1 \mu_\mathrm{ph}}, \tag{7}$$

which is the photospheric pressure scale height of the primary if it would exactly fill its Roche lobe ($R = R_\mathrm{R}$). $\mathcal{R}$ is the gas constant and μ_ph the mean molecular weight in the photosphere. We note that the phase of optically thick mass transfer ($R > R_\mathrm{R}$) ends (at point U2 in Fig. 2) only about $3\,10^6$ yr after the onset of mass transfer. During the remaining evolution the mass transfer occurs as optically thin mass loss from the atmosphere. Thus the primary underfills its critical Roche volume and strictly speaking the binary system is (slightly) detached. On the other hand, the system is detached in the classical sense, i.e. $-\Delta R \gg H_\mathrm{p}$, during its evolution form point G to I (see Figs. 2, 3, 4 and 5). This detached phase occurs because at point G the hydrogen shell source reaches and subsequently burns through a step in the hydrogen profile that in turn was left earlier by the outer convection zone (see KKW for details).

The close similarity of Figs. 3 and 5a is easily explained as a consequence of optically thin mass transfer. As is seen from Eq. (A6), log $\dot{M}$ (plotted in Fig. 3) is directly proportional to $\Delta R / H_\mathrm{p}$ (shown in Fig. 5).

5. Discussion

5.1. *Why are the results of sequences A and B so similar?*

As has become clear from the foregoing section there are no significant differences between the results of sequence A (using our new method) and sequence B (using the standard technique).

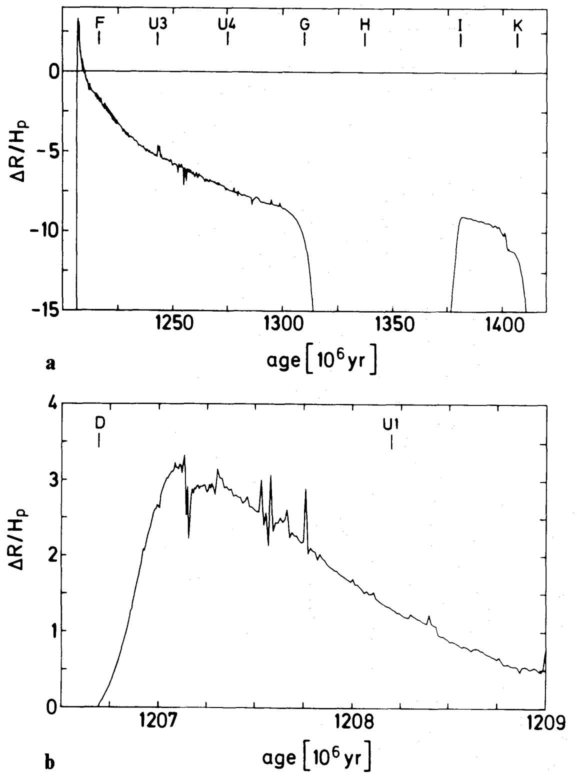

Fig. 5. **a** The difference $\Delta R := R - R_{\rm R}$ between the primary's stellar radius R and Roche radius $R_{\rm R}$ in units of the pressure scale height $H_{\rm p}$ defined in Eq. (7) as a function of time for sequence A. Note that the phase of optically thick mass transfer ($R \geq R_{\rm R}$) lasts only about 3 10^6 yr. **b** Enlarged section of diagram **a**, emphasizing the phase of optically thick mass transfer

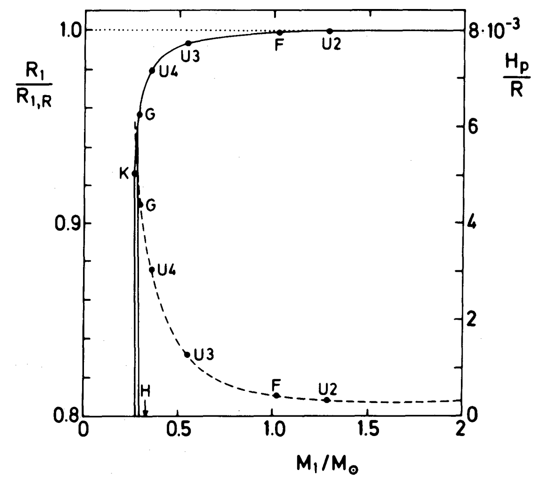

Fig. 6. The variation of $R/R_{\rm R}$ (full line) and of $H_{\rm p}/R$ (dashed line) as a function of the primary's mass. Note that most of the mass is transferred while $0.99 \lesssim R/R_{\rm R} \lesssim 1.00$ and $H_{\rm p}/R \lesssim 10^{-3}$

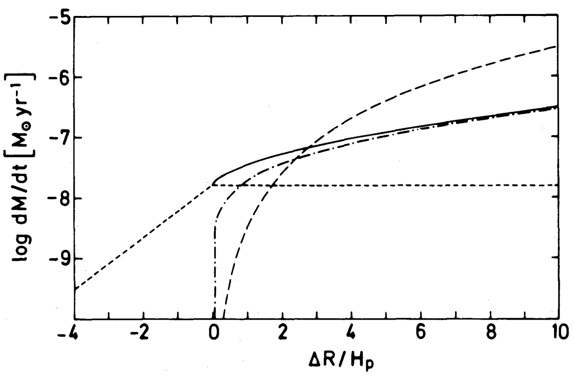

Fig. 7. Illustration of the dependence of the mass transfer rate $\dot{M}$ on $\Delta R := R - R_{\rm R}$ [given in units of $H_{\rm p}$ as defined in Eq. (7)] for a 2.0 $M_\odot$ star in a late phase of central hydrogen burning (the model is between points B and C on the evolutionary track shown in Fig. 2). For $\Delta R < 0$ only optically thin mass transfer contributes, for $\Delta R > 0$ the total mass transfer rate (solid line) is the sum of optically thin (short dashes) and thick (dash-dotted) mass transfer. The curve drawn in long dashes represents the rate of optically thick mass transfer obtained when a polytropic approximation for the outer layers of the star (polytropic index $n_2 = 1.5$, $\Gamma_1 = 5/3$) is used

The only noticeable difference is a slightly higher luminosity of the primary after the end of mass transfer (point K in Fig. 2) in sequence B. The difference of $\Delta \log L = 4\,10^{-2}$ is, however, very small and results from an even smaller difference in the mass of the degenerate helium core of $\Delta M_{\rm c} \approx 3\,10^{-3} M_\odot$, in full agreement with the core mass luminosity relation (e.g. Webbink et al., 1983). Although, at the beginning of our experiment, we had of course hoped to obtain good agreement between the results of sequences A and B, we are still surprised by how closely they match quantitatively. Therefore, we should ask ourselves why this is so. The main reason is that the mass transfer rate as given by Eqs. (A1) and (A17) depends sensitively on $\Delta R/H_{\rm p}$. This is clearly seen in Fig. 7, where we plot $\dot{M}$ as a function of $\Delta R/H_{\rm p}$ for a particular model, and also from Eq. (A6). As a consequence of this $|\Delta R/H_{\rm p}|$ never becomes very large during the mass transfer phase, i.e. in our example $-10 \lesssim \Delta R/H_{\rm p} \lesssim 3$, as can be seen from Fig. 5. Now, in our example, $H_{\rm p}/R$ is very small ($\sim 3\,10^{-4}$) at the beginning (point D) and still small ($\sim 6\,10^{-3}$) at the end of mass transfer (point K). Because of this, the boundary condition imposed on the primary in the standard technique [Eq. (1)] is also fulfilled to a good approximation in sequence A. Contrary to what one might expect naïvely from the results shown in Fig. 5, most of the mass is actually transferred when the primary's radius is very close to its Roche radius. This is shown in Fig. 6. In the same figure we show also the run of $H_{\rm p}/R$. It is seen that during the major part of the mass transfer phase $H_{\rm p}/R$ and thus $\Delta R/R$ remain very small. It is only during the latest phases of mass transfer, when it is driven by nuclear evolution, that the primary underfills its Roche lobe noticeably.

5.2. *When does the standard technique yield acceptable results?*

Having now understood, why sequences A and B are so similar, we wish to generalize our results and to answer the above question. From what we inferred from Sect. 5.1, Eq. (1) will be reasonably fulfilled whenever the following two conditions hold: (1) The mass transfer rate is a very sensitive function of ΔR, and (2) the relative pressure/density scale height H/R in the outer layers of the donor is small, i.e. $H/R \ll 1$. If these two conditions hold and if the phases of turning on or off mass transfer, during which Eq. (2) is clearly violated, are not of importance, the standard technique will yield acceptable results. Fortunately, all these conditions are fulfilled in practically all of the classical studies of close binary evolution. However, as we have already mentioned in the introduction, there are cases where one or more of the above conditions are violated, as e.g. in the late case C mass transfer studied by PR.

5.3. *Numerical problems*

Here we wish to discuss briefly our experience made with the two different numerical techniques to compute $\dot{M}$. In the standard technique, the stellar model has to be computed with Eq. (1) as an additional outer boundary condition. As a consequence, the mass M of the donor becomes a variable of the problem whose value must be determined simultaneously with the internal structure by the Henyey iteration procedure. Accordingly the transfer rate $\dot{M}$ is computed as the difference between the masses of two successive models, divided by the time step Δt. In contrast in our new method M has a fixed value during the iteration process: it is determined before starting the iteration by adding $\Delta M = \dot{M}\Delta t$ to the mass of the preceding model, where $\dot{M}$ is an explicit and very sensitive function of stellar and binary parameters. As a direct consequence of this, higher numerical accuracy is required which, among other things, means also smaller time steps. In fact, when compared to sequence B, a typical time step in sequence A was shorter by about a factor of 10 during the optically thin mass transfer phase and by about a factor 100 during optically thick mass transfer. Moreover, since $\dot{M}$, computed with the new techniques, has the tendency to oscillate around a mean value, unless the time step is chosen very short (see DMR for a discussion of this point), we had to introduce a damping algorithm to allow for reasonably long time steps. This was done by averaging the transfer rate computed according to Eqs. (A1) and (A17) over the two preceding time steps. The time scale of this damping is thus much shorter than the time scale of the oscillations visible in Figs. 3 and 5 (note that no damping was used in sequence B).

Another problem we encountered in our computations were sudden jumps in $\dot{M}$. These occur because the stellar evolution program we used integrates atmospheres and subatmospheres over a triangular grid of ($T_{\rm eff}$, L) values (see Kippenhahn et al., 1967) to determine the surface boundary conditions of a particular model. Therefore, when the evolutionary track in the HRD crosses the boundary of two adjacent triangles, jumps in R and thus in $\dot{M}$ result occasionally. We should emphasize that the standard technique is even more susceptible to such discountinuities than the new one: occasionally, after having crossed the boundary between two triangles, it becomes very difficult to find a new model with the required radius. Worse, it is often necessary to stop the computations at such a point and they can be resumed only after some tedious manual intervention. Because of this, the basic advantage of the standard technique, namely allowing for relatively long time steps and thus requiring fewer models to perform a given evolution, is lost. In the end, we find that although the new technique requires more time steps, the total amount of time going into a simulation is comparable to that taken by the standard technique.

The obvious way to remedy the above-mentioned numerical problems associated with the new technique and thus to lower the required computing time is to reformulate it in such a way (e.g. following Savonije, 1978) that M (and thus $\dot{M}$) becomes a variable of the stellar structure problem.

6. Summary and conclusions

The purpose of this paper was to examine to what extent the results of numerical calculations of classical binary evolution, using an algorithm based on Eqs. (1) and (2) for computing the mass transfer rate, hold when checked against computations using a physically superior method of determining the mass transfer rate. For this we first derived explicit relations for the mass transfer rate as a function of known stellar and binary parameters. We take properly into account the finite atmospheric scale height of the donor (as described by Ritter, 1988), and in cases where the donor overfills its critical Roche volume the mass transfer rate is determined by using the numerically computed stratification of the donor rather than a polytropic approximation, as Paczyński and Sienkiewicz (1972) and Savonije (1978) did.

Subsequently we computed the evolution of one particular binary system (with initial parameters close to those used by KKW, see Table 1), using on the one hand our improved method (sequence A) and the standard technique which involves Eqs. (1) and (2) (sequence B) on the other for computing mass transfer. Comparing the results of the two sequences (see Figs. 2, 3 and 4) we do not find any significant differences. This despite the fact that the primary in sequence A underfills its critical Roche lobe during most of the mass transfer phase (see Fig. 5). The explanation for the good agreement is that although the primary underfills its critical Roche volume by as much as 10 pressure scale heights, most of the mass is transferred when the primary's radius is actually quite close to its Roche radius (see Fig. 6). This is because the atmospheric pressure scale height is so small (see Fig. 6) that the boundary conditions used in the standard technique [Eqs. (1) and (2)] are well approximated during most of the transfer phase.

Generalizing our results and those obtained by PR and DMR one can now safely say that the standard technique used in so many classical studies of binary evolution yields acceptable results whenever the following conditions are fulfilled:

(1) The mass transfer rate is a sensitive function of the degree of over- or under-filling the Roche lobe, i.e. of $\Delta R = R - R_{\rm R}$,

(2) the density/pressure scale height H in the outer layers of the donor is small compared to the stellar radius, i.e. $H/R \ll 1$, and

(3) the phases of turning on or off mass transfer are not important.

Although the above conditions are met in probably all the classical studies of binary evolution, there are interesting exceptions where the standard method would fail (see e.g. Savonije, 1979; PR; DMR or Kolb and Ritter, in preparation).

Despite the fact that our improved method of computing mass transfer requires considerable shorter time steps and thus more models to be computed than the standard technique and despite the fact that the latter seems to work satisfactorily in most cases of interest, we prefer our improved method because it is easier to handle and, after all, less time consuming. Nevertheless, the numerical problems associated with this technique (see also DMR for this point) call for a reformulation in such a way that the mass (and thus the mass transfer rate) becomes a variable of the stellar structure problem.

Appendix: Computing the mass transfer rate

A.1. Preliminary remarks

In the following we describe the binary in a corotating Cartesian coordinate system and take its origin at the primary's centre of gravity. The z-axis is perpendicular to the orbital plane, the x-axis directed towards the secondary's centre of gravity. The plane defined by $x=x_{\rm L}$, where $(x_{\rm L}, 0, 0)$ is the inner Lagrangian point L_1, is referred to as L_1-plane. All coordinates are given in units of the orbital separation a.

A.2. Optically thin mass transfer ($R \le R_{\rm R}$)

Following Ritter (1988) the mass transfer rate is given by

$$-\dot{M}_1 = \dot{M}_0 \exp\left\{-\frac{\Phi_{\rm L}-\Phi_{\rm ph}}{\mathcal{R}T_{\rm eff}/\mu_{\rm ph}}\right\}, \tag{A1}$$

where

$$\dot{M}_0 = \frac{2\pi}{\sqrt{e}} F_1(q) \frac{R_{\rm R}^3}{GM_1}\left(\frac{\mathcal{R}T_{\rm eff}}{\mu_{\rm ph}}\right)^{3/2} \rho_{\rm ph} \tag{A2}$$

$$F_1(q) = q f_1^{-3}(q)\{g(q)[g(q)-1-q]\}^{-1/2} \tag{A3}$$

$$g(q) = \frac{q}{x_{\rm L}^3} + \frac{1}{(1-x_{\rm L})^3} \tag{A4}$$

and

$$f_1(q) = \frac{R_{\rm R}}{a}. \tag{A5}$$

Here $q=M_1/M_2$ is the mass ratio, $\rho_{\rm ph}$, $\Phi_{\rm ph}$ and $\mu_{\rm ph}$ are respectively the density, the Roche potential and the mean molecular weight in the primary's photosphere.

For evaluating $\Delta\Phi = \Phi_{\rm L} - \Phi_{\rm ph}$ one can expand the Roche potential in the vicinity of $\Phi_{\rm L}$ in terms of the volume equivalent radius (see Ritter, 1988). This yields (for small $\Delta\Phi$)

$$\Delta\Phi = \Phi_{\rm L} - \Phi_{\rm ph} \approx \frac{R_{\rm R}-R}{\hat{H}_{\rm p}}\left(\frac{\mathcal{R}T_{\rm eff}}{\mu_{\rm ph}}\right) \tag{A6}$$

where

$$\hat{H}_{\rm p} = H_{\rm p}/\gamma(q). \tag{A7}$$

$H_{\rm p}$ is the photospheric pressure scale height as defined by Eq. (7) and $\gamma(q)$ a factor that takes into account the non-sphericity of the Roche equipotential surfaces (see Ritter, 1988). For the q values of interest here $\gamma(q)$ is close to unity. An approximation can be found in Ritter (1988). (Note, however, that there the mass ratio is the inverse of our q because mass loss from the secondary is considered there.)

Yet, for large $\Delta\Phi$ the above expansion is not adequate. In this case it is better to evaluate $\Delta\Phi$ directly by interpolating an inverted form of the tables computed by Mochnacki (1984), where the stellar volume radius is given in terms of the mass ratio and a fill-out factor F defined as

$$F = \Phi_{\rm L}/\Phi_{\rm ph}. \tag{A8}$$

whence

$$\Delta\Phi = \Phi_{\rm L}\left(1+\frac{1}{F}\right). \tag{A9}$$

A.3. Optically thick mass transfer ($R > R_{\rm R}$)

As described in Sect. 2, Bernoulli's equation [Eq. (4)] relates the density $\rho_{\rm L}$ in the L_1-plane to the density in a layer at the same potential Φ of a stellar model in hydrostatic equilibrium. From Eqs. (5) and (6) and the definition of the adiabatic sound velocity

$$v_{\rm s}^2 = \Gamma_1 \frac{P}{\rho} \tag{A10}$$

we get

$$\rho_{\rm L} = \left(\frac{2}{\Gamma_1+1}\right)^{\frac{1}{\Gamma_1-1}} \rho_0. \tag{A11}$$

Applying in the following the expansion to second order of the Roche potential Φ in the L_1-plane near $\Phi=\Phi_{\rm L}$ (e.g. Savonije, 1978)

$$\Phi - \Phi_{\rm L} \approx By^2 + Cz^2, \tag{A12}$$

where

$$B = \left(\frac{\partial^2\Phi}{\partial y^2}\right)_{\rm L_1} = \frac{GM_1}{2aq}\{g(q)-1-q\}, \tag{A13}$$

$$C = \left(\frac{\partial^2\Phi}{\partial z^2}\right)_{\rm L_1} = \frac{GM_1}{2aq} g(q), \tag{A14}$$

the integral appearing in Eq. (3) can be rewritten in terms of Φ. In the approximation (A12) the intersection of an equipotential surface with the L_1-plane is an ellipse with an area

$$A = \pi\sqrt{\frac{\Phi-\Phi_{\rm L}}{B}}\sqrt{\frac{\Phi-\Phi_{\rm L}}{C}}\, a^2. \tag{A15}$$

Consequently

$$dA = \frac{\pi a^2}{\sqrt{BC}}\, d\Phi. \tag{A16}$$

Combining Eqs. (A10), (A11) and (A16) with Eq. (3) finally yields

$$-\dot{M}_1 = \dot{M}_0 + 2\pi F_1(q) \frac{R_{\rm R}^3}{GM_1}\int_{\Phi_{\rm L}}^{\Phi_{\rm ph}} F_3(\Gamma_1)\left(\frac{\mathcal{R}T_0}{\mu_0}\right)^{1/2} \rho_0\, d\Phi. \tag{A17}$$

$$F_3(\Gamma_1) = \Gamma_1^{1/2}\left(\frac{2}{\Gamma_1+1}\right)^{\frac{\Gamma_1+1}{2(\Gamma_1-1)}} \tag{A18}$$

is of order unity. The first term in (A17), given by Eq. (A2), is due to saturated optically thin mass loss, the second one contains an integral over those layers of the stellar model which are at a potential $\Phi_{\rm L} < \Phi < \Phi_{\rm ph}$ (i.e. over the optically thick layers lying outside the critical Roche potential). In our study, this integral is evaluated numerically using the stratification, i.e. Γ_1, T_0, μ_0, ρ_0, as given by the stellar model. For this it is useful to change the integration variable by using the equation of hydrostatic equilibrium (which holds in our stellar models)

$$d\Phi = -\frac{dP_0}{\rho_0}. \tag{A19}$$

In order to illustrate the typical dependence of $\dot{M}$ on the degree of under- or overfill of the Roche lobe, we show in Fig. 7 $\dot{M}$ as a function of $\Delta R/H_{\rm p}$ for one particular model (a 2 $M_\odot$ star near the end of central hydrogen burning) as computed according to Eqs. (A1) and (A17), respectively. In addition, Fig. 7 shows also the result one obtains if a polytropic approximation for the outer stellar layers is used.

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